Metallicrab - Assessing how different pollutants affect the immunological responses of Hairy Shore Crabs (*Hemigrapsus oregonensis*)

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Background - Significance

- The hairy shore crab (Hemisgrapsus oregonensis) is a common intertidal species that inhabits estuaries and coastal environments along the west coast of North America, where pollutants such as trace metals tend to accumulate (Trustcott et al., 1990)
- Estuarine habitats act as sinks for heavy metals (cadmium, lead, copper, etc.), increasing the likelihood of bioaccumulation in benthic organisms like shore crabs. (Castro-Elenes et al. 2021)
- Hemocytes, the primary immune cells in crustaceans, are highly sensitive to environmental stressors and are used as indicators of physiological responses to pollution. (Impellitteri et al., 2022)

Background - What’s Been Done?

Alimin, Ade Wahyul Fajri. (2025). Hemolymph as a biomarker to access the health of decapod crustaceans: a review. International Aquatic Research. 16. 327-342. 10.22034/iar.2024.2008915.1736

- Decapoda have open circulatory system that allows hemolymph to circulate between vascular structures and interorgan sinuses
- Hemolymph is the primary fluid for transport within the body
- Hemolymph contains two copper atoms for oxygen binding
- Hemolymph plays an important role in oxygen and nutrient distribution, metabolism, osmotic regulation, innate immune, and inflammatory responses
- There are 3 types of hemocytes: hyaline cells, semigranular cells, and granular cells
    - They can be identified through morphological characteristics
    - Each performs different functions
- Hyaline cell: phagocytosis, hemolymph coagulation, wound repair
- Semigranular cell: encapsulation, storage, and release of the porPO system, cytotoxicity, phagocytosis, wound repair
- Granular cell: cytotoxicity, storage, and release of the proPO system
- The cellular immune response is mediated by the hemocytes

Background - What's Been Done?

Truscott, R., & White, K. N. (1990). The Influence of Metal and Temperature Stress on the Immune System of Crabs. *Functional Ecology*, *4*(3), 455. https://doi.org/10.2307/2389609

- There has been an increase in disease prevalence in marine invertebrates from polluted environments --> thus a decrease in immunocompetence
- Lack of knowledge as to what pollutant is causing this
- Internal self defense mechanisms are cellular --> phagocytosis and encapsulation
- 3 types of hemocytes: hyalinocytes, semigranulocytes, granulocytes
- Pollutant effects on hemocyte numbers and phagocytic activity
- Used copper, cadmium, and mercury
    - Copper is toxic at high concentrations
    - Cadmium and mercury are toxic at low concentrations, and are not regulated
- Results: THC and DHC (differential hemocyte count) changed during and after exposure --> but a relatively small effect
    - Copper -> inhibitory effect on phagocytosis
    - Cadmium -> stimulatory effect on phagocytosis

Research Question & Hypotheses

- **Question**: How does pollution affect the immunological responses of *Hemigrapsus oregonensis*?

- **Hypotheses**:
  - H0: There will be no change in hemocyte count across all groups.
  - HA: There will be an impact on the total hemocyte count (THC) in the trace metal experimental groups.
    - HA1: THC will increase
    - HA2: THC will decrease

Manipulated Variables & Observed Responses

Pollution (trace metals)

- Types of metals
  - Lead
  - Copper
  - Steel
- Represent commonly found heavy metal pollutants that accumulate along the coast
- Metals can disrupt cellular processes and induce stress

Observed physiological responses

- Hemocyte Count
  - Indicator of immune system activity
  - Measured as total hemocytes per unit volume
- Increase in THC → immune activation (stress response)
- Decrease in THC → immune suppression or damage

Link to Hypothesis

- No change → supports H_0
- Increase → supports HA_1 (immune activation)
- Decrease → supports HA_2 (immune suppression)

Experimental Design & Sketch

Week 1

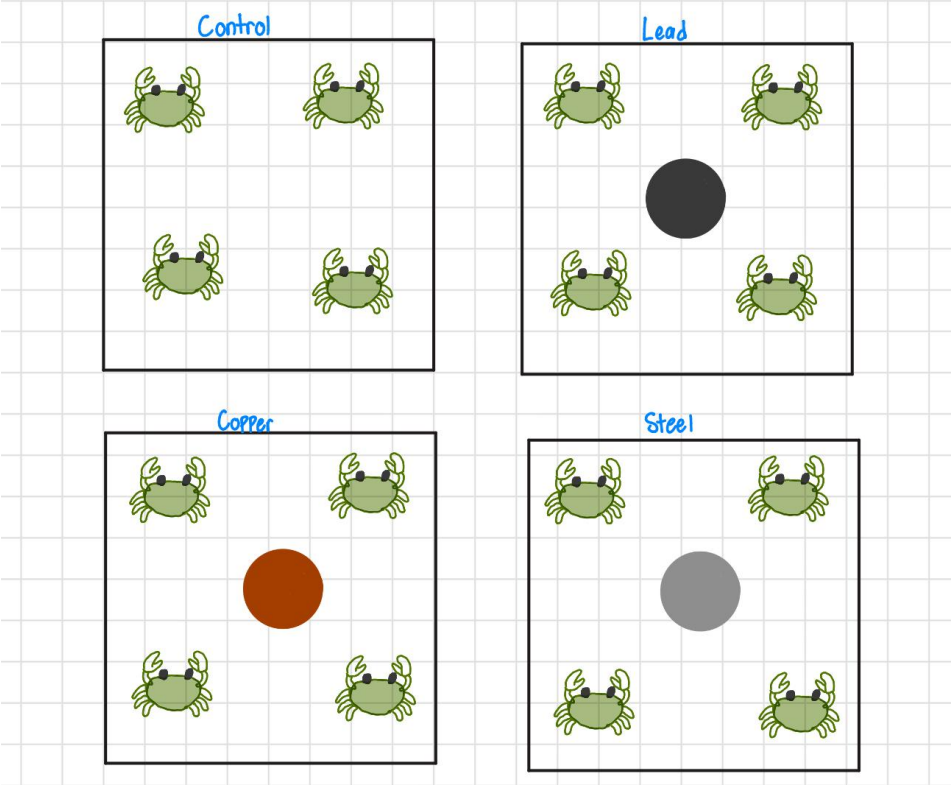
- Random selection of large Hairy Shore Crabs (*Hemigrapsus oregonensis*)
- Body measurements
- Righting time
- Collect hemolymph sample from 2 unused crabs in experiment to establish baseline
    - 1/8th of a mL
- Staining to identify different hemocytes

Week 2

- Righting time
- Collect hemolymph sample from 2 crabs in each group
- Hemocyte count & identification

Week 3

- Righting time
- Collecting hemolymph sample from 2 **different** crabs in each group
- Hemocyte count & identification



Expected Results

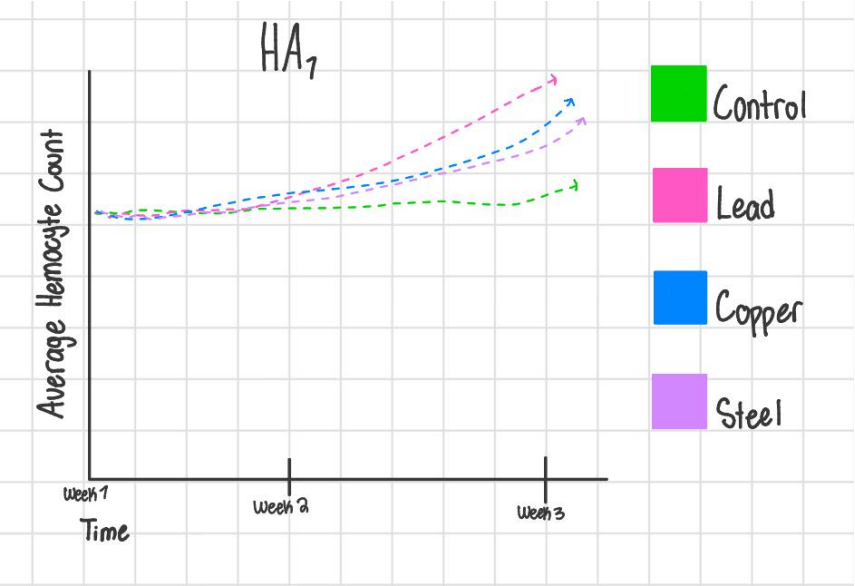


Figure 1. Expected average hemocyte count if HA1 is correct. Increase in average hemocyte count across all groups, control group having the least amount of change

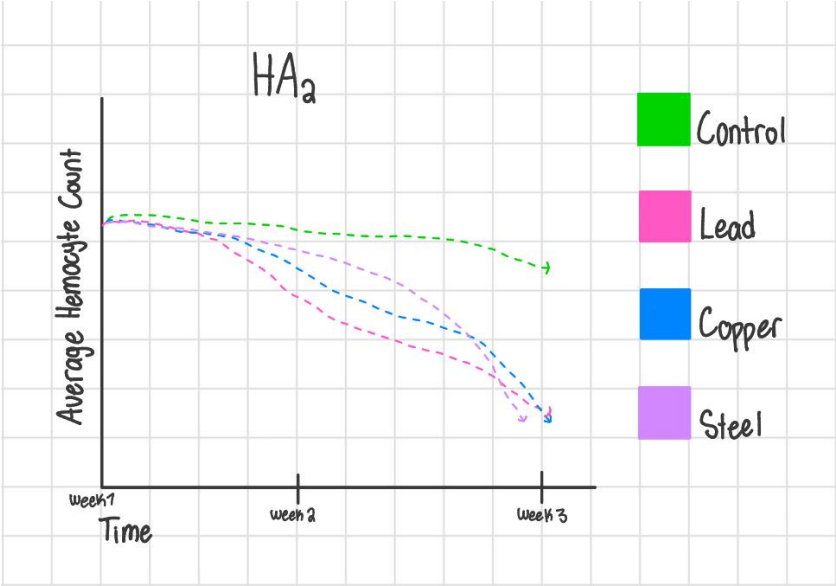


Figure 2. Expected average hemocyte count if HA2 is correct. Decrease in average hemocyte count across all groups, control group having the least amount of change

Expected Results

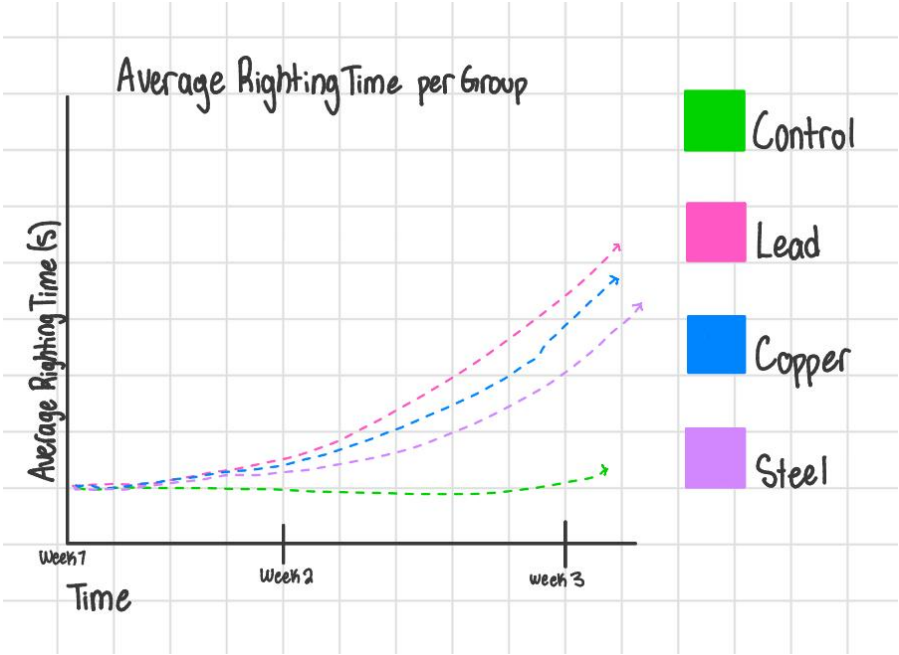


Figure 3. Expected average righting time per group over time. Increased righting time seen due to an increase in stress

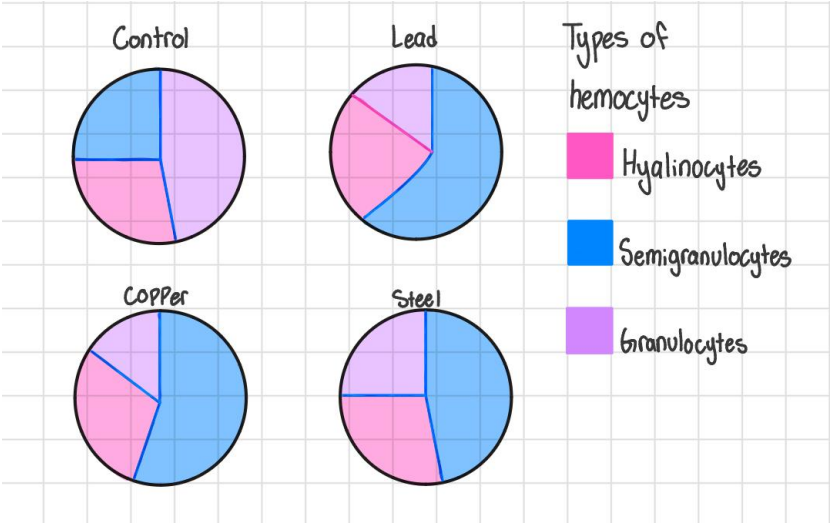


Figure 4. Expected amounts of hemocytes in each treatment group. Semigranulocytes shown to be the most dominant type of hemocyte due to its ability to encapsulate foreign materials (Qin et al., 2012)

Sources

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- Impellitteri, F., Curpăn, A.-S., Plăvan, G., Ciobica, A., & Faggio, C. (2022, December 15). *Hemocytes: A useful tool for assessing the toxicity of microplastics, heavy metals, and pesticides on aquatic invertebrates*. MDPI. https://doi.org/10.3390/ijerph192416830
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